

How to mount qcow2/vdi disk image on Linux

Last updated on November 14, 2020 by Dan Nanni

Question: I have a `qcow2`-type disk image which is used by one my guest VMs on QEMU/KVM. I want to modify the content of the disk image without powering on the VM, and for this I need to mount the disk image somewhere. Is there a way to mount a `qcow2` disk image on Linux?

When you run a guest virtual machine (VM) on a hypervisor, you create one or more disk images dedicated to the VM. As a *virtual* disk volume, a disk image represents the content and structure of a storage device (e.g., a hard disk drive or flash drive) attached to the VM. If you want to modify files in a VM's disk image without powering on the VM, you could *mount* the disk image. You then would be able to modify the content of the disk image before unmounting it.

In Linux, there are ways to mount a disk image, and different types of disk images require different approaches. If you are using `qcow2`-type disk images (used by QEMU/KVM), there are at least two ways to mount them on Linux.

Method One: `qemu-nbd`

Another method to mount a `qcow2` disk image is via `qemu-nbd`, a command-line tool that exports a disk image as a network block device (`nbd`).

You can install `qemu-nbd` on Linux as follows.

On Debian-based system:

```
xterm
$ sudo apt-get install qemu-utils
```

On Red Hat-based system:

```
xterm
$ sudo yum install qemu-img
```

To mount a `qcow2` disk image, first export the image to `nbd` as follows.

```
xterm
$ sudo modprobe nbd max_part=64
$ sudo qemu-nbd --connect=/dev/nbd0 /path/to/qcow2/image
```

The first command loads the `nbd` kernel module. The `max_part=N` option specifies the maximum number of partitions we want to manage with `nbd`. The second command exports the specified disk image as a network block device (`/dev/nbd0`). As a network block device, you can use `/dev/nbd0`, `/dev/nbd1`, `/dev/nbd2`, etc. whichever is unused. As for the disk image, make sure to specify its *full* path.

For example, to export the image `xenserver.qcow2` as `/dev/nbd0`:

```
xterm
$ sudo qemu-nbd --connect=/dev/nbd0 /var/lib/libvirt/images/xenserver.qcow2
```

After this, existing disk partitions in the disk image will be mapped to `/dev/nbd0p1`, `/dev/nbd0p2`, `/dev/nbd0p3`, etc.

To check the list of `nbd`-mapped partitions, use `fdisk`:

```
xterm
$ sudo fdisk /dev/nbd0 -l
```

Finally, choose any one partition (e.g., `/dev/nbd0p1`) and mount it to a local mount point (e.g., `/mnt`).

```
xterm
$ sudo mount /dev/nbd0p1 /mnt
```

Now you will be able to access and modify the content of the disk image's mounted partition via `/mnt` mount point.

Once you are done, unmount it, and disconnect the disk image as follows.

xterm

```
$ sudo umount /mnt
```

```
$ sudo qemu-nbd --disconnect /dev/nbd0
```

```
dan@poweredge:~$
```

Export disk image to network block device

```
dan@poweredge:~$
```

```
dan@poweredge:~$ sudo qemu-nbd --connect=/dev/nbd0 /var/lib/libvirt/images/xenserver.qcow2
```

```
dan@poweredge:~$
```

```
dan@poweredge:~$ sudo fdisk /dev/nbd0 -l
```

```
Disk /dev/nbd0: 12 GiB, 12884901888 bytes, 25165824 sectors
```

```
Units: sectors of 1 * 512 = 512 bytes
```

```
Sector size (logical/physical): 512 bytes / 512 bytes
```

```
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disklabel type: gpt
```

```
Disk identifier: 518988BD-A6C5-4C66-AFAB-825D2D7F5244
```

Available devices in the disk image

Device	Start	End	Sectors	Size	Type
/dev/nbd0p1	2048	8388641	8386594	4G	Microsoft basic data
/dev/nbd0p2	8390656	16777249	8386594	4G	Microsoft basic data
/dev/nbd0p3	16779264	25165790	8386527	4G	Linux LVM

```
dan@poweredge:~$
```

```
dan@poweredge:~$ sudo mount /dev/nbd0p1 /mnt
```

Mount one device in the disk image

```
dan@poweredge:~$
```

```
dan@poweredge:~$
```

```
dan@poweredge:~$ ls /mnt
```

bin	dev	home	lost+found	opt	root	srv	tmp
boot	etc	lib	media	proc	sbin	sys	usr
cli-rt	EULA	lib64	mnt	Read_Me_First.html	selinux	tftpboot	var

```
dan@poweredge:~$
```

```
dan@poweredge:~$
```

```
dan@poweredge:~$ sudo umount /mnt
```

Unmount the device

```
dan@poweredge:~$
```

```
dan@poweredge:~$
```